Online Incident Response Planning under Model Misspecification through Bayesian Learning and Belief Quantization

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Abstract

Effective responses to cyberattacks require fast decisions, even when information about the attack is incomplete or inaccurate. However, most decision-support frameworks for incident response rely on a detailed system model that describes the incident, which restricts their practical utility. In this paper, we address this limitation and present an online method for incident response planning under model misspecification, which we call MOBAL: Misspecified Online Bayesian Learning. MOBAL iteratively refines a conjecture about the model through Bayesian learning as new information becomes available, which facilitates model adaptation as the incident unfolds. To determine effective responses online, we quantize the conjectured model into a finite Markov model, which enables efficient response planning through dynamic programming. We prove that Bayesian learning is asymptotically consistent with respect to the information feedback. Additionally, we establish bounds on misspecification and quantization errors. Experiments on the CAGE-2 benchmark show that MOBAL outperforms the state of the art in terms of adaptability and robustness to model misspecification.

CCS Concepts

• **Security and privacy → Network security**.

Keywords

Cybersecurity, reinforcement learning, Bayesian learning, POMDP, misspecification, incident response, network security.

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1 Introduction

Incident response refers to the coordinated actions taken to contain, mitigate, and recover from cyberattacks. In practice, incident response is largely a manual process carried out by security experts. Although effective in many cases, this approach is slow, resource-intensive, and requires substantial expertise. For example, a recent study reports that organizations take an average of 73 days to respond and recover from an incident [30]. Reducing this delay

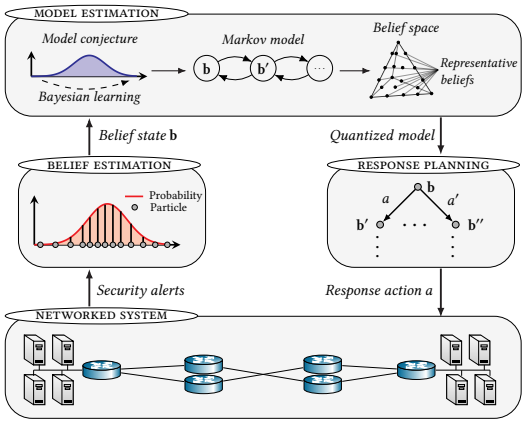


Figure 1: Our method (MOBAL) for incident response planning under model misspecification. At each time step, we estimate a belief about the system's security state and use it to update a conjecture about the system model through Bayesian learning. We then use this conjecture to sample a specific Markov model, whose belief space is quantized. Finally, we use the quantized model to efficiently compute an effective response through dynamic programming.

requires better decision-support systems to assist operators during incident handling. Currently, the standard approach to assisting operators relies on *response playbooks* [52], which comprise predefined rules for handling specific incidents. However, playbooks still rely on security experts for configuration and are therefore difficult to keep aligned with evolving threats and system architectures [47].

To address these drawbacks, significant research efforts have started to develop tools for automating the computation of effective incident response strategies for networked systems. This research draws on concepts and methods from various fields, most notably control theory [21], game theory [3, 38], dependability [63], large language models (LLMs) [15, 16, 37, 45], and reinforcement learning [27, 42, 58]. Broadly speaking, the approach in this line of research is to first construct a model or simulator of the system and then compute an optimal response strategy using numerical methods, such as dynamic programming [22], reinforcement learning [58], tree search [20], or LLMs [16, 19]. As a consequence, the quality of the resulting response strategy depends critically on the accuracy of the model or simulator, which must capture the system's (causal) *dynamics*, i.e., how the system evolves in response to attacks and response actions. However, such accurate models and simulators are generally not available in practice due to the complexity of operational systems and the uncertainty about attacks [17]. Hence, the practical applicability of current solutions is limited.

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In this paper, we address this limitation by presenting an online method for incident response planning under *model misspecification*, which we call MOBAL: Misspecified Online Bayesian Learning; see Fig. 1. In particular, we relax the standard assumption that the system model is known and only assume a probabilistic *conjecture* about the model, which may be misspecified in the sense that it assigns 0 probability to the true model. In our method, this conjecture is iteratively adapted based on available threat information via Bayesian learning. We then use the updated conjecture to compute an effective response strategy using dynamic programming.

A key challenge when performing this computation is the complexity of the dynamic programming problem, which results from two factors: (*i*) the system's security state is only partially observable; and (*ii*) the number of possible system states is large and typically grows exponentially with the system's size. As a consequence, effective incident response requires planning over a high-dimensional *belief space*, i.e., a space of probability distributions over possible states. To address this computational complexity, our method *quantizes* the belief space of the conjectured model, which enables efficient computation of a near-optimal response strategy.

We prove that MOBAL converges to a conjectured model that is consistent with the observed threat data. Moreover, we derive bounds on both the approximation error (due to quantization) and the misspecification error. To evaluate MOBAL experimentally, we apply it to CAGE-2 [14], which is a standard benchmark to evaluate incident response frameworks. The results show that MOBAL offers substantial improvements in adaptability and robustness to model misspecification compared to the state-of-the-art methods.

We summarize our contributions as follows:

- We develop MOBAL, an online method for incident response planning under model misspecification. It involves a novel combination of Bayesian learning and belief quantization.
- We derive theoretical bounds on both the approximation error introduced by the quantization and the error due to model misspecification. We also quantify the interplay between these two errors and establish conditions under which the conjectured model learned by MOBAL converges.
- We evaluate MOBAL on CAGE-2 [14], which involves responding to an advanced persistent threat in an IT infrastructure. The results show that MOBAL outperforms the state-of-the-art in adaptability and robustness to model misspecification.

2 Use Case

We consider a general incident response use case that involves the IT infrastructure of an organization. The operator of this infrastructure, which we call the *defender*, takes measures to protect it against an *attacker* while providing services to a client population. An example infrastructure is shown in Fig. 2. This infrastructure is segmented into zones with interconnected servers, which clients access through a public gateway. Though intended for service delivery, this gateway is also accessible to a potential attacker who aims to compromise servers. To achieve this goal, the attacker can perform various actions, such as reconnaissance, brute-force attacks, lateral movement, and exploits (i.e., cyber kill chain [28, 36]).

Given these attacker capabilities, we study the problem of developing *optimal incident response strategies* that map infrastructure statistics to automated actions for mitigating potential attacks while minimizing service disruption. Examples of response actions include shutdown, access control, and network resegmentation.

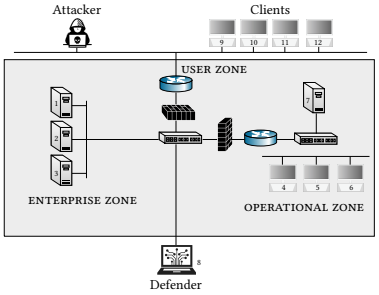


Figure 2: The actors and systems involved in the incident response use case. The system configuration and topology correspond to the CAGE-2 system [14].

3 Formalizing the Incident Response Use Case

We formulate the incident response use case described above as a partially observable Markov decision process (POMDP). Following this formalism, a response strategy π is a function that sequentially prescribes *response actions* $a_0, a_1, \ldots$ based on a series of *observations* $o_1, o_2, \ldots$ (e.g., system metrics). These actions stochastically influence the evolution of the system's *state* s_t, which captures its security and service status. Due to limited monitoring capabilities or intentional concealment by a potential attacker, the state of the system cannot be observed directly. Therefore, response actions are selected based on a *belief state* $\mathbf{b}_t$, which represents the conditional probability distribution over possible states of the system given observations. The effectiveness of these actions is quantified through a *cost function* that should be minimized.

We denote the set of response actions by $\mathcal{A}$, the set of observations by $\mathcal{O}$, and the set of states by $\mathcal{S} = \{1, \ldots, n\}$, all of which are finite. State transitions $s \to s'$ under action a occur at discrete times t according to transition probabilities $p_{ss'}(a)$. Each transition is associated with a real-valued cost $c(s, a)$ and an observation o, which is generated with probability $z(o \mid s')$. While the POMDP involves imperfect state information, it can be reformulated as an equivalent problem with perfect state information; see e.g., [64]. In this formulation, the system is described by the belief state $\mathbf{b} = \big(\mathbf{b}(1), \mathbf{b}(2), \ldots, \mathbf{b}(n)\big)$, where $\mathbf{b}(i)$ is the conditional probability that the state is i, given the history of actions and observations. This vector belongs to the belief space $\mathcal{B}$ and is updated as

$$\mathbf{b}_t = \mathbb{B}(\mathbf{b}_{t-1}, a_{t-1}, o_t), \tag{1}$$

where $\mathbb{B}$ is a given belief estimator.

We adopt the belief-space formulation and consider response strategies π that map the belief space $\mathcal{B}$ to the action space $\mathcal{A}$. Our goal is to minimize the expected discounted cumulative cost, i.e.,

$$\underset{\pi \in \Pi}{\text{minimize}} \lim_{T \to \infty} \mathop{\mathbb{E}}_{(s_t, \mathbf{b}_t)_{t \geq 0}} \left\{ \sum_{t=0}^{T} \gamma^t c(s_t, \pi(\mathbf{b}_t)) \mid \mathbf{b}_0 \right\}, \tag{2}$$

where Π is the strategy space, $\mathbb{E}\{\cdot\}$ denotes the expectation operator, s_t is the state at time t, and $\gamma \in (0, 1)$ is a discount factor. We say that a strategy $\pi^\star$ is optimal if it achieves this minimization. Such

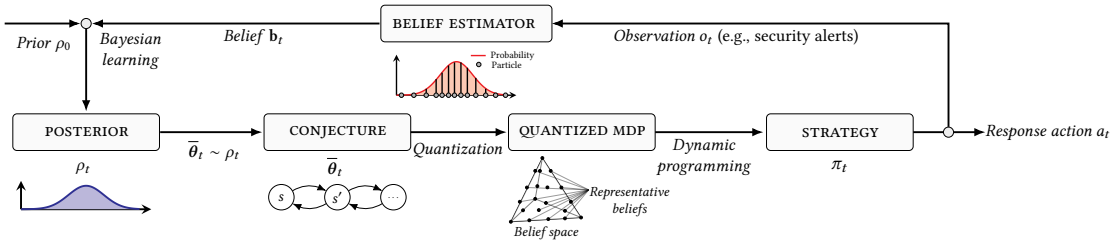


Figure 3: MOBAL: an iterative method for online learning of incident response strategies under model misspecification. The figure illustrates a time step during which (*i*) the posterior distribution over possible system models is updated via Bayesian learning based on feedback from the system; (*ii*) a conjectured model is sampled from the posterior and quantized into a computationally tractable MDP; and (*iii*) a response strategy is computed using dynamic programming.

a strategy is related to the optimal cost function $J^\star$ through the Bellman equations

$$\pi^\star(\mathbf{b}) \in \arg\min_{a\in\mathcal{A}} \left[\hat{c}(\mathbf{b},a) + \gamma \sum_{\mathbf{b}'\in\mathcal{B}} p_{\boldsymbol{\theta}}(\mathbf{b}' \mid \mathbf{b},a) J^\star(\mathbf{b}')\right], \tag{3a}$$

$$J^\star(\mathbf{b}) = \min_{a\in\mathcal{A}} \left[\hat{c}(\mathbf{b},a) + \gamma \sum_{\mathbf{b}'\in\mathcal{B}} p_{\boldsymbol{\theta}}(\mathbf{b}' \mid \mathbf{b},a) J^\star(\mathbf{b}')\right], \tag{3b}$$

$$\hat{c}(\mathbf{b},a) = \sum_{s=1}^{n} \mathbf{b}(s) c(s,a), \tag{3c}$$

where $p_{\boldsymbol{\theta}}(\mathbf{b}' \mid \mathbf{b}, a)$ is the probability of transitioning from belief $\mathbf{b}$ to belief $\mathbf{b}'$ when taking action a. We assume that the transition probabilities are parameterized by a parameter vector $\boldsymbol{\theta}$. Since the transition probabilities depend on the attacker's behavior, we consider the vector $\boldsymbol{\theta}$ to be unknown and assume only a probabilistic *conjecture* about $\boldsymbol{\theta}$, which we express through a probability distribution ρ_t over some set Θ of plausible parameter vectors. We say that the conjecture distribution ρ_t is *misspecified* if $\boldsymbol{\theta} \notin \Theta$.

Remark 1. *Since the state, action and observation spaces are assumed finite, it follows that a) an optimal strategy exists; and b) for each belief* $\mathbf{b}$ *and action* a*, the transition probability* $p_{\boldsymbol{\theta}}(\mathbf{b}' \mid \mathbf{b}, a)$ *is non-zero only for a finite set of beliefs* $\mathbf{b}'$*; see e.g., [33, Thms. 7.6.1–7.6.2] for details. Hence, the Bellman equations in (3) are well-defined.*

4 Misspecified Online Bayesian Learning

Building on the preceding problem formulation, we develop an online method for incident response planning that accounts for misspecification, which we call MOBAL: Misspecified Online Bayesian Learning. Our method evolves through a sequence of iterative steps $t = 0, 1, 2, \ldots$, as illustrated in Fig. 3. Each step includes three stages. First, we use the observations $o_1, \ldots, o_t$ (e.g., security alerts) to estimate a belief $\mathbf{b}_t$ about the system state through the belief estimator (1). Second, we use the same observations to update the distribution ρ_t and conjecture the parameter vector $\boldsymbol{\theta}$ [cf. (3)] as $\overline{\boldsymbol{\theta}} \sim \rho_t$. Lastly, we use the conjecture $\overline{\boldsymbol{\theta}}$ to construct a computationally tractable Markov decision process (MDP) via belief quantization, which allows us to efficiently approximate an optimal incident response strategy through dynamic programming. These three stages are formally defined next, starting with belief estimation.

4.1 Belief Estimation

In the context of incident response, the belief state represents a probabilistic estimate of the system's security state, which encodes information about services and possible attacks. Consequently, accurate belief estimation is key to making informed response decisions amidst uncertainty about potential attacks.

The belief state at time t can be computed via the recursion

$$\mathbf{b}_t(s') = \frac{z(o_t \mid s') \sum_{s=1}^{n} \mathbf{b}_{t-1}(s) p_{ss'}(a_{t-1})}{\sum_{i=1}^{n}\sum_{j=1}^{n} z(o_t \mid j) \mathbf{b}_{t-1}(i) p_{ij}(a_{t-1})}, \quad \text{for all } s' \in \mathcal{S}. \tag{4}$$

However, the complexity of this calculation is quadratic in the number of states n, which can become computationally intractable for systems with large state spaces. In such cases, the belief state can be efficiently estimated through *particle filtering* as

$$\widehat{\mathbf{b}}_t(s) = \frac{1}{M} \sum_{j=1}^{M} \delta_{s\widehat{s}_t^{(j)}}, \qquad \text{for all } s \in \mathcal{S}, \tag{5}$$

where $\delta_{ij} = 1$ if $i = j$, $\delta_{ij} = 0$ if $i \neq j$, and $\widehat{s}_t^1, \ldots, \widehat{s}_t^M$ are states (particles) sampled with probability proportional to the numerator in (4). Such sampling ensures that the estimated belief $\widehat{\mathbf{b}}_t$ converges (almost surely) to $\mathbf{b}_t$ when $M \to \infty$. Hence, the particle filter provides a consistent way to estimate beliefs while allowing computational complexity to be adjusted by tuning the number of particles M.

4.2 Bayesian Learning of the System Model

Given the updated belief state, the second step of MOBAL is to refine the conjecture about the system model based on the observation o_t. Specifically, we update the conjecture distribution ρ_t (treated as the probability density function) according to

$$\rho_t(\overline{\boldsymbol{\theta}}) = \frac{P(o_t \mid \overline{\boldsymbol{\theta}}, \mathbf{b}_{t-1}, a_{t-1}) \rho_{t-1}(\overline{\boldsymbol{\theta}})}{\int_\Theta P(o_t \mid \boldsymbol{\theta}', \mathbf{b}_{t-1}, a_{t-1}) \rho_{t-1}(\boldsymbol{\theta}') \mathrm{d}\boldsymbol{\theta}'} \quad \text{for all } \overline{\boldsymbol{\theta}} \in \Theta, \tag{6}$$

where $P(o_t \mid \overline{\boldsymbol{\theta}}, \mathbf{b}_{t-1}, a_{t-1})$ is the probability of the observation o_t conditioned on the conjectured parameter vector $\overline{\boldsymbol{\theta}}$, the belief state $\mathbf{b}_{t-1}$, and the response action a_{t-1}. The goal when refining the conjecture distribution ρ_t in this way is to concentrate probability density on parameter vectors $\overline{\boldsymbol{\theta}} \sim \rho_t$ that are consistent with the observations $o_1, \ldots, o_t$. In other words, we seek to minimize the *discrepancy* between the observation distribution in the (conjectured) model parameterized by $\overline{\boldsymbol{\theta}}$ and the true model parameterized

by $\boldsymbol{\theta}$; cf. (3). We define this discrepancy as

$$K(\overline{\boldsymbol{\theta}}, \nu_t) = \mathbb{E}_{\mathbf{b}\sim\nu_t}\mathbb{E}_o\left\{\ln\left(\frac{P(o \mid \boldsymbol{\theta}, \mathbf{b})}{P(o \mid \overline{\boldsymbol{\theta}}, \mathbf{b})}\right) \mid \boldsymbol{\theta}, \mathbf{b}, \pi\right\}, \tag{7}$$

where $P(o \mid \boldsymbol{\theta}, \mathbf{b})$ is obtained by marginalizing $P(o \mid \boldsymbol{\theta}, \mathbf{b}, a)$ using the empirical action distribution based on the actions $a_0, a_1, \ldots, a_{t-1}$. Similarly, ν_t denotes the empirical belief distribution[1], i.e.,

$$\nu_t(\mathbf{b}) = \frac{1}{t}\sum_{\tau=1}^{t}\delta_{\mathbf{b}\mathbf{b}_\tau}, \qquad \text{for all } \mathbf{b} \in \mathcal{B}.$$

We say that a conjecture $\overline{\boldsymbol{\theta}}$ that minimizes the discrepancy K [cf. (7)] is *consistent* [9, 55]. Hence, the set of consistent conjectures at time t is given by

$$\Theta^\star(\nu_t) = \arg\min_{\overline{\boldsymbol{\theta}}\in\Theta} K(\overline{\boldsymbol{\theta}}, \nu_t). \tag{8}$$

A desirable property of the posterior ρ_t [cf. (6)] is that it concentrates on the consistent conjectures $\Theta^\star(\nu_t)$. This property is guaranteed asymptotically under suitable conditions, as stated below.

Proposition 1 (Consistent conjectures). *Under suitable regularity conditions (see Appendix B), the following holds*

$$\lim_{t\to\infty}\int_\Theta \left(K(\overline{\boldsymbol{\theta}}, \nu_t) - K^\star_\Theta(\nu_t)\right)\rho_{t+1}(\overline{\boldsymbol{\theta}})\mathrm{d}\overline{\boldsymbol{\theta}} = 0 \text{ almost surely,} \tag{9}$$

where $K^\star_\Theta(\nu_t)$ is a finite constant defined as

$$K^\star_\Theta(\nu_t) = \min_{\overline{\boldsymbol{\theta}}\in\Theta} K(\overline{\boldsymbol{\theta}}, \nu_t).$$

While Prop. 1 ensures that the posterior ρ_t eventually concentrates on consistent conjectures [cf. (8)], it does not quantify how close the dynamics induced by these conjectures are to the true system dynamics. In particular, if the true parameter vector $\boldsymbol{\theta}$ lies outside the set Θ, then even the most consistent conjecture may yield a transition model $p_{\overline{\boldsymbol{\theta}}}(\mathbf{b}' \mid \mathbf{b}, a)$ that deviates significantly from the true model $p_{\boldsymbol{\theta}}(\mathbf{b}' \mid \mathbf{b}, a)$. As a result, a response strategy derived from such a conjecture may be suboptimal. To formalize this suboptimality, let $\overline{J}^\star$ denote the optimal cost function in the model defined by $\overline{\boldsymbol{\theta}}$; cf. (3b). We refer to the difference between this cost function and the optimal cost function $J^\star$ [cf. (3b)] as the *misspecification error*. This error is bounded by the difference between $p_{\overline{\boldsymbol{\theta}}}$ and $p_{\boldsymbol{\theta}}$, as stated in the following proposition.

Proposition 2 (Misspecification error bound). *If the transition probability distributions $p_{\boldsymbol{\theta}}$ and $p_{\overline{\boldsymbol{\theta}}}$ satisfy*

$$\sum_{\mathbf{b}'\in\mathcal{B}} |p_{\boldsymbol{\theta}}(\mathbf{b}' \mid \mathbf{b}, a) - p_{\overline{\boldsymbol{\theta}}}(\mathbf{b}' \mid \mathbf{b}, a)| \le \alpha, \text{ for all } \mathbf{b} \in \mathcal{B}, a \in \mathcal{A}, \tag{10}$$

for some constant $\alpha \in [0, 2]$. Then we have

$$\|\overline{J}^\star - J^\star\|_\infty \le \frac{\gamma\alpha c_{MAX}}{(1-\gamma)^2},$$

where γ is the discount factor and c_{MAX} is a finite constant defined by

$$c_{MAX} = \max_{\mathbf{b}\in\mathcal{B}, a\in\mathcal{A}} \hat{c}(\mathbf{b}, a). \tag{11}$$

This proposition quantifies the cost of relying on a misspecified model. It states that the misspecification error grows proportionally with the error of the conjectured state transitions; cf. (10).

[1]We use the standard convention that $-\ln 0 = \infty$ and $0 \ln 0 = 0$.

4.3 Model Quantization and Response Planning

Given the updated belief and conjecture, the last step of MOBAL is to compute an effective response strategy. While an optimal strategy (according to the conjecture $\overline{\boldsymbol{\theta}}$) can (in principle) be computed using dynamic programming techniques, this computation is intractable due to the continuous belief space $\mathcal{B}$. To circumvent this intractability, we *quantize* $\mathcal{B}$ into a finite set of representative beliefs. Specifically, we define the set of representative beliefs as

$$\tilde{\mathcal{B}} = \left\{\tilde{\mathbf{b}} \;\middle|\; \tilde{\mathbf{b}} \in \mathcal{B}, \tilde{\mathbf{b}}(s) = \frac{\beta_s}{r}, \sum_{s\in\mathcal{S}}\beta_s = r, \beta_s \in \{0, \ldots, r\}\right\}, \tag{12}$$

where $r \in \{1, 2, \ldots\}$ is a given parameter that can be interpreted as the *quantization resolution*. To relate the beliefs $\mathbf{b} \in \mathcal{B}$ to the representative beliefs $\tilde{\mathbf{b}} \in \tilde{\mathcal{B}}$, we define a mapping $\Phi : \mathcal{B} \mapsto \tilde{\mathcal{B}}$ as

$$\Phi(\mathbf{b}) = \arg\min_{\tilde{\mathbf{b}}\in\tilde{\mathcal{B}}} \|\mathbf{b} - \tilde{\mathbf{b}}\|_\infty, \qquad \text{for all } \mathbf{b} \in \mathcal{B}, \tag{13}$$

where ties in the argmin are broken in a consistent way. Given this mapping from the belief space $\mathcal{B}$ to the set of representative beliefs $\tilde{\mathcal{B}}$, we obtain a well-defined MDP whose state space is the set of representative beliefs $\tilde{\mathcal{B}}$. The cost function in this MDP is given by (3c) and the transition probabilities are defined as

$$\hat{p}_{\overline{\boldsymbol{\theta}}}(\tilde{\mathbf{b}}' \mid \tilde{\mathbf{b}}, a) = \sum_{\mathbf{b}'\in\mathcal{B}} p_{\overline{\boldsymbol{\theta}}}(\mathbf{b}' \mid \tilde{\mathbf{b}}, a)\delta_{\tilde{\mathbf{b}}'\Phi(\mathbf{b}')}, \quad \text{for all } a \in \mathcal{A}, \tilde{\mathbf{b}}', \tilde{\mathbf{b}} \in \tilde{\mathcal{B}}.$$

Due to the finite state space, the quantized MDP can be efficiently solved using dynamic programming. Let $V^\star$ and $\mu^\star$ denote the optimal cost function and strategy in this MDP, respectively. Similarly, let $\overline{J}^\star$ and $\overline{\mu}^\star$ denote the optimal cost function and strategy of the (non-quantized) POMDP based on the conjecture $\overline{\boldsymbol{\theta}}$, respectively; cf. (3). We can then approximate $\overline{J}^\star$ and $\overline{\mu}^\star$ as

$$\tilde{J}(\mathbf{b}) = V^\star(\Phi(\mathbf{b})) \text{ and } \tilde{\pi}(\mathbf{b}) = \mu^\star(\Phi(\mathbf{b})), \text{ for all } \mathbf{b} \in \mathcal{B}. \tag{14}$$

We refer to the difference between the cost function approximation $\tilde{J}$ and the (conjectured) optimal cost function $\overline{J}^\star$ as the *approximation error*. To understand this error, note that the mapping Φ [cf. (13)] partitions the belief space $\mathcal{B}$ into disjoint subsets as

$$\mathcal{B} = \bigcup_{\tilde{\mathbf{b}}\in\tilde{\mathcal{B}}} S_{\tilde{\mathbf{b}}}, \qquad \text{where } S_{\tilde{\mathbf{b}}} = \left\{\mathbf{b} \mid \mathbf{b} \in \mathcal{B}, \Phi(\mathbf{b}) = \tilde{\mathbf{b}}\right\}. \tag{15}$$

In view of (14), this partitioning means that the approximation error of $\tilde{J}$ is determined by how much the (conjetured) optimal cost function $\overline{J}^\star(\mathbf{b})$ varies for beliefs $\mathbf{b}$ within the same belief-space partition $S_{\tilde{\mathbf{b}}}$. This insight is formalized by the following proposition.

Proposition 3 (Approximation error bound). *The error of the cost function approximation $\tilde{J}$ [cf. (14)] with respect to the conjectured optimal cost function $\overline{J}^\star$ is bounded as*

$$|\tilde{J}(\mathbf{b}) - \overline{J}^\star(\mathbf{b})| \le \frac{\epsilon}{1-\gamma}, \qquad \text{for all } \mathbf{b} \in S_{\tilde{\mathbf{b}}}, \tilde{\mathbf{b}} \in \tilde{\mathcal{B}},$$

where γ is the discount factor and ϵ is a finite constant defined by

$$\epsilon = \max_{\tilde{\mathbf{b}}\in\tilde{\mathcal{B}}} \sup_{\mathbf{b},\mathbf{b}'\in S_{\tilde{\mathbf{b}}}} |\overline{J}^\star(\mathbf{b}) - \overline{J}^\star(\mathbf{b}')|. \tag{16}$$

The meaning of Prop. 3 is that the error of the cost function approximation $\tilde{J}$ [cf. (14)] is small if the aggregation mapping Φ [cf. (13)] conforms to the (conjectured) optimal cost function $\overline{J}^\star$ in the sense that Φ varies little in regions of the belief space where $\overline{J}^\star$ also varies little. This error can be controlled by tuning the quantization resolution r [cf. (12)], as stated in the following proposition.

Proposition 4 (Asymptotic (conjectured) optimality). *Given the cost function approximation $\tilde{J}$ [cf. (14)], the following holds.*

$$\lim_{r\to\infty} |\tilde{J}(\mathbf{b}) - \overline{J}^\star(\mathbf{b})| = 0, \qquad \text{for all } \mathbf{b} \in \mathcal{B}.$$

While the preceding propositions quantify the difference between the cost function approximation $\tilde{J}$ [cf. (14)] and the *conjectured* optimal cost function $\overline{J}^\star$, they do not say anything about the difference to the optimal cost function $J^\star$. This difference depends on both the approximation error $\|\overline{J}^\star - \tilde{J}\|_\infty$ and the misspecification error $\|\overline{J}^\star - J^\star\|_\infty$, as captured by the following theorem.

Theorem 1 (Sub-optimality bound of MOBAL). *The sub-optimality of the cost function approximation $\tilde{J}$ [cf. (14)] is bounded as*

$$\|\tilde{J} - J^\star\|_\infty \leq \frac{\epsilon}{1-\gamma} + \frac{\gamma\alpha c_{MAX}}{(1-\gamma)^2},$$

where γ is the discount factor and $(\epsilon, \alpha, c_{MAX})$ are the finite constants defined in (16), (10), and (11), respectively.

Proof. By Prop. 3, we have □

$$\|\tilde{J} - \overline{J}^\star\|_\infty \leq \frac{\epsilon}{1-\gamma},$$

and by Prop. 2, we have

$$\|\overline{J}^\star - J^\star\|_\infty \leq \frac{\gamma\alpha c_{\text{MAX}}}{(1-\gamma)^2}.$$

Applying the triangle inequality, we obtain

$$\|\tilde{J} - J^\star\|_\infty \leq \|\tilde{J} - \overline{J}^\star\|_\infty + \|\overline{J}^\star - J^\star\|_\infty \leq \frac{\epsilon}{1-\gamma} + \frac{\gamma\alpha c_{\text{MAX}}}{(1-\gamma)^2}.$$

This theorem shows that the sub-optimality of MOBAL decomposes into two components: one due to model approximation (ϵ) and one due to model misspecification (α). It is significant because it shows that performance guarantees can be obtained even when relaxing the standard assumption of a correctly specified model.

Summary of our method for incident response (Fig. 1)

MOBAL starts with an initial conjecture ρ_0 about the incident and a belief $\mathbf{b}_0$ about the security state. Given these priors, MOBAL proceeds through a sequence of iterative steps $t = 0, 1, 2, \ldots$, where each step consists of three stages.

(1) The belief $\mathbf{b}_t$ is updated based on the latest observation o_t through recursive state estimation; cf. (5).

(2) The conjecture distribution ρ_t is adapted to the observation o_t through Bayesian learning; cf. (6).

(3) A conjecture is sampled $\overline{\theta}_t \sim \rho_t$ and used to approximate an optimal response action a_t through dynamic programming and belief quantization; cf. (14).

5 Illustrative Example

To illustrate our method for incident response planning under model misspecification, we consider the response scenario introduced in [22]. This scenario consists of a networked system with N components; see Fig. 4. Each component has two states: 1 (compromised) or 0 (safe), i.e., $s = (s^1, \ldots, s^N)$ where $s^l \in \{0, 1\}$. Compromises occur randomly over time and incur operational costs. Intrusion detection systems generate observations $o = (o^1, \ldots, o^N)$ that provide partial indications of the components' states, where $o^l \in \{0, 1, \ldots\}$ is the number of security alerts related to component l. The security policy π prescribes the action vector $a = (a^1, \ldots, a^N)$, where each a^l determines whether to block network traffic to component l ($a^l = 1$) or take no action ($a^l = 0$). Blocking a component can prevent further compromise or lateral movement by an attacker, but may also disrupt legitimate services. The goal is to determine a response strategy that balances this trade-off optimally.

We capture this objective through the cost function

$$c(s, a) = \sum_{l=1}^{N} \overbrace{2s^l(1 - a^l)}^{\text{intrusion cost}} + \overbrace{a^l}^{\text{blocking cost}}, \tag{17}$$

which encodes that costs are incurred for unmitigated intrusions ($s^l = 1$) and for blocking network traffic ($a^l = 1$).

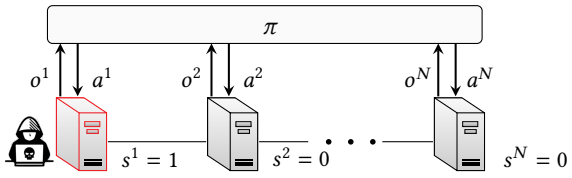


Figure 4: The system in the illustrative example [22].

We define the observation distribution for each component using the Beta-binomial distribution shown in Fig. 5. Specifically, we define the distribution of o^l as BetaBin(7, 1, 0.7) when $s^l = 1$ and define the distribution as BetaBin(7, 0.7, 3) when $s^l = 0$. These distributions reflect that alerts may occur ($o^l > 0$) during normal operation ($s^l = 0$) but are more likely during attacks ($s^l = 1$). Similar alert distributions have been observed in practice; see e.g., [23].

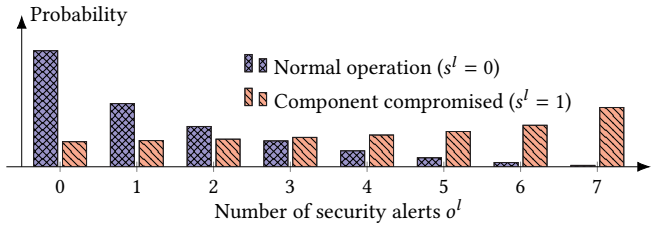


Figure 5: Observation distribution per component l in the illustrative example.

The transition probabilities $p_{ss'}(a)$ are defined as follows. If component l is compromised ($s^l = 1$), then it remains so until recovery is applied ($a^l = 1$), at which point the state s^l is set to 0. Otherwise, the probability that it becomes compromised is $\min\{p_A(1 + \mathcal{N}_l(s)), 1\}$,

where $\mathcal{N}_l(s)$ is the number of compromised neighbors of component l in the network and $p_A \in (0, 1]$ is a given parameter. This compromise probability reflects how attacks can propagate through neighboring components in the network.

For the numerical examples presented in the following, we consider the case where all parameters of the model are known except p_A, which we define as $p_A = 0.2$. We define the initial conjecture distribution of this parameter to be a uniform distribution over the set $\Theta = \{0, 0.5, 1\}$, i.e., $\rho_0(0) = \rho_0(0.5) = \rho_0(1) = \frac{1}{3}$; cf. (6).

Numerical examples. We start by evaluating the accuracy of the particle filter (5). Figure 6 shows the difference between the estimated belief and the true belief. As expected, the accuracy improves with the number of particles M. For small systems (e.g., $N = 3$ components), we find that the particle filter provides a close approximation to the true belief with only $M = 10$ particles.

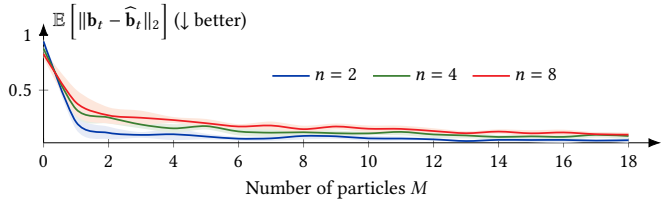


Figure 6: Expected error of the particle filter for the illustrative example in function of the number of particles M for varying sizes of the state space n (corresponding to $N = 1$, $N = 2$, and $N = 3$ system components); cf. (5). Curves show the mean value from evaluations with 100 random seeds; shaded areas indicate standard deviations and $\|\cdot\|_2$ denotes the Euclidean norm. The belief $\mathbf{b}_t$ is calculated using formula (4) with $\overline{\theta} = \theta$. We calculate the expectation by running 100 POMDP episodes of 100 time steps each with strategy $\tilde{\pi}$ [cf. (14)] computed using quantization resolution $r = 5$.

Now consider the Bayesian learning formula (6). Figure 7 shows the evolution of the posterior ρ_t [cf. (6)] and the discrepancy K [cf. (7)]. We observe that the posterior ρ_t converges to a distribution that concentrates on the conjecture $\overline{\theta} = 0$, which is the conjecture with the lowest discrepancy, as expected from Prop. 1.

Next, we analyze how close the bound in Prop. 3 is to the actual approximation error, i.e., the difference $\|\tilde{J} - \overline{J}^\star\|_\infty$. Figure 8 shows that the bound is not tight but becomes increasingly accurate as the resolution r increases, as asserted in Prop. 3. However, increasing r also causes the number of representative beliefs to grow, which is illustrated in Fig. 9. Hence, r governs a trade-off between computational expedience and approximation error.

Figure 10 shows the structure of the optimal cost function $J^\star$, the conjectured optimal cost function $\overline{J}^\star$, and the cost function approximation $\tilde{J}$; cf. (14). We observe that $J^\star$ and $\overline{J}^\star$ have a similar structure but differ significantly in their values. Moreover, we observe that $\tilde{J}$ is piece-wise constant, as expected from (13).

6 Evaluation on the CAGE-2 Benchmark

To compare our method with the state-of-the-art methods for computing incident response strategies, we apply it to the CAGE-2 benchmark [14]. CAGE-2 involves a networked system segmented into

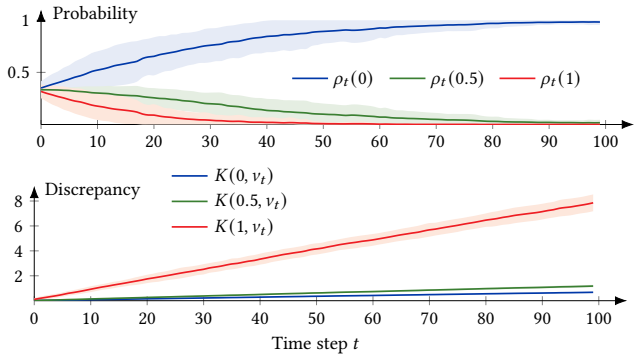


Figure 7: Evolution of the posterior conjecture distribution ρ_t [cf. (6)] and the discrepancy $K(\overline{\theta}, \nu_t)$ [cf. (7)] for the illustrative example. In this example, the true parameter vector is $\theta = 0.2$, the set of conjectures is $\Theta = \{0, 0.5, 1\}$, and the quantization resolution is $r = 5$.

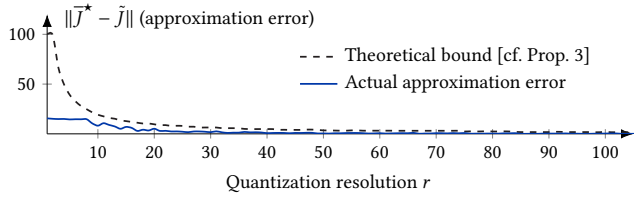


Figure 8: Comparison between the theoretical error bound in Prop. 3 and the actual error of the cost function approximation $\tilde{J}$ [cf. (14)] for the illustrative example with $N = 1$ and varying quantization resolutions r; cf. (12).

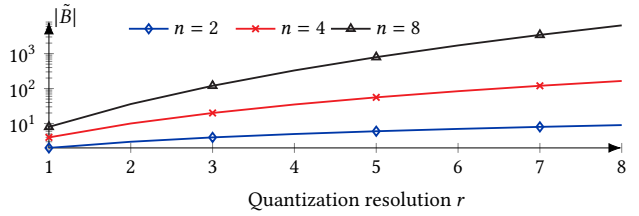


Figure 9: Number of representative beliefs [cf. (12)] in function of the quantization resolution; curves relate to state spaces of different sizes.

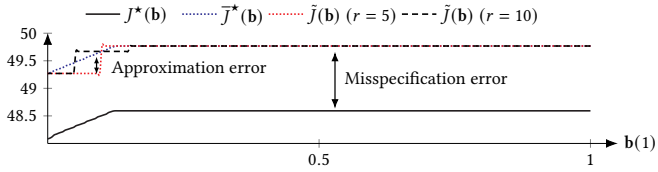


Figure 10: Comparison between the optimal cost function $J^\star$, the optimal cost function in the conjectured model $\overline{J}^\star$, and the approximation $\tilde{J}$ with varying quantization resolution r [cf. (14)] for the illustrative example. The curves for the approximations are computed using $\overline{\theta} = 0.5$ and $\theta = 0.2$. The number of system components in the example is $N = 1$. Hence, the value $\mathbf{b}(1)$ on the x-axis indicates the belief of system compromise.

zones with servers and workstations that run network services. The network topology of the CAGE-2 system is shown in Fig. 2. The system provides services to clients through a gateway, which is also open to an attacker who aims to intrude on the system. These services generate a stream of network statistics, which are input to an incident response strategy π, which can take four *actions* on each node: analyze it for a possible intrusion; start a decoy service; remove malware; and restore it to a secure state, which temporarily disrupts its service. Each service disruption and node compromise incurs a predefined cost; the problem is finding a response strategy that minimizes this cost. When formulated as a POMDP, CAGE-2 has 145 actions, over 10^{47} states, and over 10^{25} observations [20].

Experimental setup. We consider the standard CAGE-2 setup with the B-LINE attacker and run the system for 100 time steps. Due to the large state and observation spaces in CAGE-2, we employ the following approximations to instantiate MOBAL. First, we use the particle filter (5) with $M = 50$ particles to implement the belief estimator (1). Second, we approximate the Bayesian update (6) using Monte-Carlo sampling. Third, since the dimension of the belief space $\mathcal{B}$ is larger than 10^{47}, we reduce its dimension by only considering beliefs over the following state variables: the attacker's state (i.e., the attacker's location in the network), the attacker's next target, and the configuration of the decoys. These state features were originally proposed in [22] and lead to a belief space of dimension 427,500, which we quantize with resolution $r = 1$; cf. (12).

Evaluation scenarios. A central aspect of a response strategy in CAGE-2 is the selection and placement of decoys, which are intended to mislead the attacker and divert attention away from vulnerable system components. The effectiveness of the decoys depends on the probability that an attacker will engage with them and the number of decoys available. Table 2 lists the attack-probabilities in CAGE-2 under different decoy configurations. As shown in the table, the probability of a successful attack decreases with the number of decoys. Current methods for CAGE-2 assume that these probabilities are encoded in a simulator that can be used for numerical optimization of the response strategy. In practice, however, these probabilities cannot be known with certainty and can only be conjectured. For this reason, we consider both the standard CAGE-2 scenario and a (more realistic) scenario in which the potential benefit of the decoys is unknown. The scenarios are detailed below.

(1) NO MISSPECIFICATION: In this scenario, we consider the case where the model is correctly specified and known, i.e., $\rho_0(\boldsymbol{\theta}) = 1$, where $\boldsymbol{\theta}$ are the true parameters of the CAGE-2 model. (The source code of CAGE-2 is available in [14].)
(2) MISSPECIFICATION: In this scenario, we consider the case where the model is misspecified, i.e., $\rho_0(\boldsymbol{\theta}) = 0$. We define the vector $\boldsymbol{\theta}$ to represent the conditional probability of a successful attack against a node given its decoy configuration. Accordingly, we define Θ to be a set of conjectures of these probabilities; see Table 2. We assume that all other parameters of the CAGE-2 model are correctly specified.

Methods for comparison. Over 35 methods have been evaluated against the CAGE-2 benchmark. We compare our method (MOBAL) against the state-of-the-art methods, namely: CARDIFF [54] and C-POMCP [20, Alg. 1]. We also compare it against two baseline methods: PPO [48, Alg. 1] and POMCP [51, Alg. 1]. For the MISSPECIFICATION scenario, we run these methods on a simulator of CAGE-2 where all of the probability parameters listed in Table 2 are fixed to 0.5.

Evaluation results. The evaluation results are summarized in Table 1. In the NO MISSPECIFICATION SCENARIO, the results show that our method (MOBAL) performs slightly worse than the state-of-the-art (C-POMCP and CARDIFF), but performs significantly better than the baseline methods (PPO and POMCP). However, in the MISSPECIFICATION scenario, MOBAL significantly outperforms all other methods. We attribute the favorable performance of MOBAL to its ability to adapt the conjectured system model online based on system observations. By contrast, the existing methods assume a correctly specified model and cannot adapt it.

Method	*Offline/Online compute time (min)*	*Cost (↓ better)*
No misspecification		
MOBAL	0/8.50	15.19 ± 0.82
CARDIFF [54]	300/0.01	**13.69** ± 0.53
PPO [48]	1000/0.01	119.02 ± 58.11
C-POMCP [20]	0/0.50	**13.32** ± 0.18
POMCP [51]	0/0.50	29.51 ± 2.00
Misspecification		
MOBAL	0/8.50	**35.91** ± 9.01
CARDIFF [54]	300/0.01	94.28 ± 33.27
PPO [48]	1000/0.01	124.38 ± 55.49
C-POMCP [20]	0/0.50	92.71 ± 27.67
POMCP [51]	0/0.50	91.51 ± 28.23

Table 1: Evaluation results on CAGE-2. Rows relate to different methods; columns indicate performance metrics. Results that are within the margin of statistical equivalence to the state-of-the-art are highlighted in bold. Numbers indicate the mean and the standard deviation from 100 evaluations with 100 time steps. The cost is calculated using CAGE-2's internal cost function.

Discussion of the evaluation results. The evaluation demonstrates the key benefit of our method (MOBAL), namely its robustness to model misspecification. While existing methods perform well when the system model is correctly specified, their reliance on a detailed model makes them brittle in practice where such models are unavailable. In contrast, MOBAL continuously adapts a conjecture about the model based on observed data, which allows it to respond effectively to attacks even under model misspecification.

7 Related Work

Since the early 1980s, there has been a broad interest in automating security functions, especially in intrusion detection and incident response [4]. Traditional methods for incident response rely on static rules that map infrastructure statistics to response actions [6, 57]. The main drawback of these methods is their dependence on domain experts to configure the rules, a process that is both labor-intensive and costly. Substantial effort has been devoted to addressing this limitation by developing methods for *automatically* computing effective incident response strategies. Three predominant approaches have emerged from this research: control-theoretic, reinforcement learning, and game-theoretic approaches.

Node	*SMTP decoy*	*TOMCAT decoy*	*APACHE decoy*	*FTP decoy*	*FEMITTER decoy*	*SMSS decoy*	*SSH decoy*	*Attack probability* θ	*Conjectures* Θ
1	✗	✗	✗	✗	✗	✗	✗	1	{0, 0.5, 1}
1	✓	✗	✗	✗	✗	✗	✗	0.25	{0, 0.5, 1}
1	✓	✓	✗	✗	✗	✗	✗	0.12	{0, 0.5, 1}
1	✓	✓	✗	✓	✗	✗	✗	0.08	{0, 0.5, 1}
1	✓	✓	✓	✓	✗	✗	✗	0.08	{0, 0.5, 1}
2	✗	✗	✗	✗	✗	✗	✗	1	{0, 0.5, 1}
2	✗	✗	✗	✗	✓	✗	✗	0.25	{0, 0.5, 1}
3	✗	✗	✗	✗	✗	✗	✗	1	{0, 0.5, 1}
3	✗	✗	✗	✗	✓	✗	✗	0.25	{0, 0.5, 1}
7	✗	✗	✗	✗	✗	✗	✗	1	{0, 0.5, 1}
7	✓	✗	✗	✗	✗	✗	✗	0.25	{0, 0.5, 1}
7	✓	✗	✓	✗	✗	✗	✗	0.13	{0, 0.5, 1}
7	✓	✓	✓	✗	✗	✗	✗	0.08	{0, 0.5, 1}
7	✓	✓	✓	✓	✗	✗	✗	0.08	{0, 0.5, 1}
9	✗	✗	✗	✗	✗	✗	✗	1	{0, 0.5, 1}
9	✗	✗	✓	✗	✗	✗	✗	0.09	{0, 0.5, 1}
9	✗	✓	✓	✗	✗	✗	✗	0.08	{0, 0.5, 1}
9	✗	✓	✓	✗	✗	✓	✗	0.08	{0, 0.5, 1}
9	✗	✓	✓	✗	✗	✓	✓	0.08	{0, 0.5, 1}
10	✗	✗	✗	✗	✗	✗	✗	1	{0, 0.5, 1}
10	✗	✗	✗	✗	✓	✗	✗	0.25	{0, 0.5, 1}
10	✗	✓	✗	✗	✓	✗	✗	0.17	{0, 0.5, 1}
10	✗	✓	✓	✗	✓	✗	✗	0.12	{0, 0.5, 1}
10	✗	✓	✓	✗	✓	✗	✓	0.10	{0, 0.5, 1}
11	✗	✗	✗	✗	✗	✗	✗	1	{0, 0.5, 1}
11	✗	✗	✗	✓	✗	✗	✗	1	{0, 0.5, 1}
11	✗	✗	✗	✓	✗	✗	✓	0.09	{0, 0.5, 1}
12	✗	✗	✗	✗	✗	✗	✗	1	{0, 0.5, 1}

Table 2: Decoy configurations and attack probabilities for the CAGE-2 system [14]. The node identifiers correspond to the identifiers shown in Fig. 2. The last column indicates the conjectured attack probabilities by our method (MOBAL). While the CAGE-2 system includes 12 nodes (including the clients and the defender), only a subset of them are amenable to host decoys, which is why not all nodes are listed in the table. Moreover, different types of decoys are compatible with different types of nodes, which is why not all decoy configurations are listed in the table; see [14] for details.

Control theory for automated incident response. Control theory provides a well-established mathematical framework for studying automatic systems. Therefore, it provides a foundational theory for automated incident response. Previous works that apply control theory to incident response in IT systems include: [23, 29, 40], all of which model incident response as the problem of controlling a discrete-time dynamical system and obtain optimal strategies through dynamic programming techniques.

Reinforcement learning for automated incident response. Reinforcement learning has emerged as a promising approach to approximate optimal control strategies in scenarios where dynamic programming is not applicable, and fundamental breakthroughs demonstrated by systems like ALPHA-GO [50] have inspired researchers to study reinforcement learning to automate security functions. Three early papers: [18], [59], and [49] analyze incident response and apply traditional reinforcement learning algorithms. They have catalyzed much follow-up research [5, 7, 8, 19, 25, 26, 34, 35, 42, 43]. These works show that *deep* reinforcement learning is a scalable technique for approximating optimal response strategies. However, such methods often lack convergence guarantees and rely on efficient simulators for training.

Game theory for automated incident response. Game theory stands out from control theory and reinforcement learning by focusing on decision-makers that *reason strategically* about the opponents' behavior. The formulation of incident response as a game can be traced back to the early 2000s with works such as [13] and [3]. In addition to these early pioneers, numerous researchers have contributed to this line of research in the last two decades; see e.g., [2, 24, 41, 56, 62]. These works study various aspects of security games, including the existence, uniqueness, and structure of equilibria, as well as computational methods. However, most of them are based on abstract models and how they generalize to complex systems like CAGE-2 is unproven.

Comparison with this paper. The main difference between this paper and the works referenced above is that we propose a method for *online* learning of incident response strategies under *model misspecification*. By contrast, virtually all referenced works are *offline* methods that assume access to a *correctly specified* system model or simulator. The advantage of our method (MOBAL) is that it applies to a much broader class of practical use cases.

The only existing method for response planning that manages model misspecification in a principled way is our earlier work [21], which has influenced aspects of MOBAL. However, the method proposed in [21] is designed for a small-scale game-theoretic setting, whereas we focus on a large-scale POMDP setting. The benefit of our approach is that it allows us to compare MOBAL against the state-of-the-art on the CAGE-2 benchmark. Another fundamental difference between MOBAL and the method proposed in [21] is the computational approach. Whereas we compute response strategies based on model quantization and dynamic programming, the method in [21] uses lookahead optimization and rollout [12].

8 Practical Considerations

The practical deployment of MOBAL depends on the characteristics of the target environment, such as network topology, system size, and response time requirements. While our experimental evaluation in this paper is focused on IT systems, our problem formulation is general and can be instantiated for a broad range of operational contexts, including on-premises, cloud-based, hybrid, and operational technology (OT) systems. Our POMDP model [cf. §3] treats the environment as a set of partially observable states, actions, and observations, without imposing restrictions on the physical infrastructure. For example, in IT systems, the observation o may represent alerts from an intrusion detection system. Similarly, in OT systems, o could capture sensor readings or control system alarms.

9 Conclusion

Effective incident response often requires quick decisions based on partial (and possibly misleading) indicators of compromise. In this paper, we address this challenge by designing a method for incident response planning that explicitly accounts for model misspecification, which we call MOBAL: Misspecified Online Bayesian Learning. Our method starts from a potentially inaccurate model conjecture and continuously adapts it using Bayesian updates informed by system observations. To compute effective responses online, we quantize this conjecture at each time step into a finite Markov model, which enables efficient response planning via dynamic programming. We establish theoretical guarantees for convergence and derive bounds that quantify the effects of model misspecification and quantization. Experiments on the CAGE-2 benchmark show that our method offers substantial improvements in robustness to model misspecification compared to the current state-of-the-art methods.

Future work. While we have evaluated MOBAL on the CAGE-2 benchmark, testing it in additional environments is an important next step. Furthermore, the current online computational time of MOBAL is around 8.5 minutes per time step. Though this planning time is acceptable in many contexts, it may be prohibitive for time-critical incident response scenarios. Future research should therefore investigate ways to reduce the planning time. To this end, a promising approach is to combine MOBAL with offline computations.

Acknowledgments

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References

[1] Charalambos D. Aliprantis and Kim C. Border. 2006. *Infinite Dimensional Analysis*. Springer Berlin, Heidelberg. doi:10.1007/3-540-29587-9

[2] Tansu Alpcan and Tamer Basar. 2010. *Network Security: A Decision and Game-Theoretic Approach* (1st ed.). Cambridge University Press, USA.

[3] Eitan Altman, Konstantin Avrachenkov, and Andrey Garnaev. 2007. A Jamming Game in Wireless Networks with Transmission Cost. In *NET-COOP*.

[4] James P. Anderson. 1980. *Computer Security Threat Monitoring and Surveillance*. Technical Report Contract 79F26400. James P. Anderson Co., Fort Washington, PA. Prepared for the United States Air Force.

[5] Andy Applebaum, Camron Dennler, Patrick Dwyer, Marina Moskowitz, Harold Nguyen, Nicole Nichols, Nicole Park, Paul Rachwalski, Frank Rau, Adrian Webster, and Melody Wolk. 2022. Bridging Automated to Autonomous Cyber Defense: Foundational Analysis of Tabular Q-Learning. In *Proceedings of the 15th ACM Workshop on Artificial Intelligence and Security*. doi:10.1145/3560830.3563732

[6] Andy Applebaum, Shawn Johnson, Michael Limiero, and Michael Smith. 2018. Playbook Oriented Cyber Response. In *2018 National Cyber Summit (NCS)*. 8–15. doi:10.1109/NCS.2018.00007

[7] Elizabeth Bates, Vasilios Mavroudis, and Chris Hicks. 2023. Reward Shaping for Happier Autonomous Cyber Security Agents. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security* (Copenhagen, Denmark) *(AISec '23)*. Association for Computing Machinery, New York, NY, USA, 221–232. doi:10.1145/3605764.3623916

[8] Yahuza Bello and Ahmed Refaey Hussein. 2024. Dynamic Policy Decision/Enforcement Security Zoning Through Stochastic Games and Meta Learning. *IEEE Transactions on Network and Service Management* (2024), 1–1. doi:10.1109/TNSM.2024.3481662

[9] Robert H. Berk. 1966. Limiting Behavior of Posterior Distributions when the Model is Incorrect. *The Annals of Mathematical Statistics* 37, 1 (1966), 51 – 58. doi:10.1214/aoms/1177699597

[10] Dimitri Bertsekas. 2019. Feature-based aggregation and deep reinforcement learning: A survey and some new implementations. *IEEE/CAA Journal of Automatica Sinica* 6, 1 (2019), 1–31. doi:10.1109/JAS.2018.7511249

[11] Dimitri Bertsekas. 2019. *Reinforcement learning and optimal control*. Athena Scientific.

[12] Dimitri Bertsekas. 2025. *A Course in Reinforcement Learning*. Athena Scientific. 2nd edition.

[13] Levente Buttyán and Hubaux Jean-Pierre. 2001. Rational Exchange - A Formal Model Based on Game Theory. In *Electronic Commerce*, Ludger Fiege, Gero Mühl, and Uwe Wilhelm (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 114–126.

[14] CAGE. 2022. TTCP CAGE Challenge 2. In *AAAI-22 Workshop on Artificial Intelligence for Cyber Security (AICS)*.

[15] Alberto Castagnaro, Mauro Conti, and Luca Pajola. 2024. Offensive AI: Enhancing Directory Brute-forcing Attack with the Use of Language Models. In *Proceedings of the 2024 Workshop on Artificial Intelligence and Security* (Salt Lake City, UT, USA) *(AISec '24)*. Association for Computing Machinery, New York, NY, USA, 184–195. doi:10.1145/3689932.3694770

[16] Sebastián R. Castro, Roberto Campbell, Nancy Lau, Octavio Villalobos, Jiaqi Duan, and Alvaro A. Cardenas. 2025. Large Language Models are Autonomous Cyber Defenders. arXiv:2505.04843 [cs.AI] https://arxiv.org/abs/2505.04843

[17] Yunfei Ge, Tao Li, and Quanyan Zhu. 2023. Scenario-Agnostic Zero-Trust Defense with Explainable Threshold Policy: A Meta-Learning Approach. In *IEEE INFOCOM 2023 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*. 1–6. doi:10.1109/infocomwkshps57453.2023.10225816

[18] James Cannady Georgia. 2000. Next Generation Intrusion Detection: Autonomous Reinforcement Learning of Network Attacks. In *In Proceedings of the 23rd National Information Systems Secuity Conference*. 1–12.

[19] Kim Hammar, Tansu Alpcan, and Emil C. Lupu. 2025. Incident Response Planning Using a Lightweight Large Language Model with Reduced Hallucination. arXiv:2508.05188 [cs.CR] https://arxiv.org/abs/2508.05188

[20] Kim Hammar, Neil Dhir, and Rolf Stadler. 2024. Optimal Defender Strategies for CAGE-2 using Causal Modeling and Tree Search. arXiv:2407.11070 [cs.LG] https://arxiv.org/abs/2407.11070

[21] Kim Hammar, Tao Li, Rolf Stadler, and Quanyan Zhu. 2025. Adaptive Security Response Strategies Through Conjectural Online Learning. *IEEE Transactions on Information Forensics and Security* 20 (2025), 4055–4070. doi:10.1109/TIFS.2025.3558600

[22] Kim Hammar, Yuchao Li, Tansu Alpcan, Emil C. Lupu, and Dimitri Bertsekas. 2025. Adaptive network security policies via belief aggregation and rollout. https://arxiv.org/abs/2507.15163.

[23] Kim Hammar and Rolf Stadler. 2024. Intrusion Tolerance for Networked Systems through Two-Level Feedback Control. In *2024 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*. 338–352. doi:10.1109/DSN58291.2024.00042

[24] Yi Han, Tansu Alpcan, Jeffrey Chan, Christopher Leckie, and Benjamin I. P. Rubinstein. 2016. A Game Theoretical Approach to Defend Against Co-Resident Attacks in Cloud Computing: Preventing Co-Residence Using Semi-Supervised Learning. *IEEE Transactions on Information Forensics and Security* 11, 3 (2016), 556–570. doi:10.1109/TIFS.2015.2505680

[25] Yi Han, Benjamin I. P. Rubinstein, Tamas Abraham, Tansu Alpcan, Olivier De Vel, Sarah Erfani, David Hubczenko, Christopher Leckie, and Paul Montague. 2018. Reinforcement Learning for Autonomous Defence in Software-Defined Networking. In *Decision and Game Theory for Security*, Linda Bushnell, Radha Poovendran, and Tamer Başar (Eds.). Springer International Publishing, Cham, 145–165.

[26] Chris Hicks, Vasilios Mavroudis, Myles Foley, Thomas Davies, Kate Highnam, and Tim Watson. 2023. Canaries and Whistles: Resilient Drone Communication Networks with (or without) Deep Reinforcement Learning. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security* (Copenhagen, Denmark) *(AISec '23)*. Association for Computing Machinery, New York, NY, USA, 91–101. doi:10.1145/3605764.3623986

[27] Zequan Huang, Jacques Robin, Nicolas Herbaut, Nourhène Ben Rabah, and Bénédicte Le Grand. 2025. Toward an Intent-Based and Ontology-Driven Autonomic Security Response in Security Orchestration Automation and Response.

arXiv:2507.12061 [cs.CR] https://arxiv.org/abs/2507.12061
[28] Eric M Hutchins, Michael J Cloppert, Rohan M Amin, et al. 2011. Intelligence-driven computer network defense informed by analysis of adversary campaigns and intrusion kill chains. *Leading Issues in Information Warfare & Security Research* 1, 1 (2011), 80.
[29] Stefano Iannucci, Qian Chen, and Sherif Abdelwahed. 2016. High-Performance Intrusion Response Planning on Many-Core Architectures. In *International Conference on Computer Communication and Networks (ICCCN)*.
[30] IBM Security and Ponemon Institute. 2024. *Cost of a Data Breach Report 2024.* Technical Report 19. IBM, Cambridge, MA. Based on breaches at 524 organizations across 17 industries in 16 countries between March 2023 and February 2024..
[31] C. T. Ionescu-Tulcea. 1949. Mesures dans les espaces produits. *Atti della Accademia Nazionale dei Lincei, Rendiconti Classe di Scienze Fisiche, Matematiche e Naturali* 7 (1949), 208–211.
[32] Michael Kearns and Satinder Singh. 2002. Near-Optimal Reinforcement Learning in Polynomial Time. *Machine Learning* 49, 2 (01 Nov 2002), 209–232. doi:10.1023/A:1017984413808
[33] Vikram Krishnamurthy. 2016. *Partially Observed Markov Decision Processes: From Filtering to Controlled Sensing.* Cambridge University Press. doi:10.1017/CBO9781316471104
[34] Mehmet Necip Kurt, Oyetunji Ogundijo, Chong Li, and Xiaodong Wang. 2019. Online Cyber-Attack Detection in Smart Grid: A Reinforcement Learning Approach. *IEEE Transactions on Smart Grid* 10, 5 (2019), 5174–5185. doi:10.1109/TSG.2018.2878570
[35] Tao Li, Kim Hammar, Rolf Stadler, and Quanyan Zhu. 2024. Conjectural Online Learning with First-order Beliefs in Asymmetric Information Stochastic Games. In *2024 IEEE 63rd Conference on Decision and Control (CDC).* 6780–6785. doi:10.1109/CDC56724.2024.10886479
[36] Tao Li, Yunian Pan, and Quanyan Zhu. 2024. Decision-Dominant Strategic Defense Against Lateral Movement for 5G Zero-Trust Multi-Domain Networks. In *Network Security Empowered by Artificial Intelligence.* Springer Nature Switzerland, Cham, Switzerland, 25–76. doi:10.1007/978-3-031-53510-9_2
[37] Tao Li, Ya-Ting Yang, Yunian Pan, and Quanyan Zhu. 2025. From Texts to Shields: Convergence of Large Language Models and Cybersecurity. *arXiv preprint arXiv:2505.00841* (2025). arXiv:2505.00841
[38] Tao Li and Quanyan Zhu. 2025. Symbiotic game and foundation models for cyber deception operations in strategic cyber warfare. In *Foundations of Cyber Deception.* Springer Cham, Cham, Switzerland. [Online] Available at https://arxiv.org/pdf/2403.10570.
[39] Yuchao Li, Kim Hammar, and Dimitri Bertsekas. 2025. Feature-Based Belief Aggregation for Partially Observable Markovian Decision Problems. https://arxiv.org/abs/2507.04646.
[40] Erik Miehling, Mohammad Rasouli, and Demosthenis Teneketzis. 2018. A POMDP Approach to the Dynamic Defense of Large-Scale Cyber Networks. *IEEE Transactions on Information Forensics and Security* 13, 10 (2018). doi:10.1109/TIFS.2018.2819967
[41] Shana Moothedath, Dinuka Sahabandu, Joey Allen, Andrew Clark, Linda Bushnell, Wenke Lee, and Radha Poovendran. 2020. A Game-Theoretic Approach for Dynamic Information Flow Tracking to Detect Multistage Advanced Persistent Threats. *IEEE Trans. Automat. Control* 65, 12 (2020), 5248–5263. doi:10.1109/TAC.2020.2976040
[42] Thanh Thi Nguyen and Vijay Janapa Reddi. 2023. Deep Reinforcement Learning for Cyber Security. *IEEE Transactions on Neural Networks and Learning Systems* 34, 8 (2023), 3779–3795. doi:10.1109/TNNLS.2021.3121870
[43] Trung V. Phan and Thomas Bauschert. 2022. DeepAir: Deep Reinforcement Learning for Adaptive Intrusion Response in Software-Defined Networks. *IEEE Transactions on Network and Service Management* (2022), 1–1. doi:10.1109/TNSM.2022.3158468
[44] Antonin Raffin, Ashley Hill, Adam Gleave, Anssi Kanervisto, Maximilian Ernestus, and Noah Dormann. 2021. Stable-Baselines3: Reliable Reinforcement Learning Implementations. *Journal of Machine Learning Research* 22, 268 (2021), 1–8. http://jmlr.org/papers/v22/20-1364.html
[45] Maria Rigaki, Ondřej Lukáš, Carlos A. Catania, and Sebastian Garcia. 2023. Out of the Cage: How Stochastic Parrots Win in Cyber Security Environments. arXiv:2308.12086 [cs.CR]
[46] Naci Saldi, Serdar Yüksel, and Tamás Linder. 2017. On the Asymptotic Optimality of Finite Approximations to Markov Decision Processes with Borel Spaces. *Math. Oper. Res.* 42, 4 (Nov. 2017), 945–978. doi:10.1287/moor.2016.0832
[47] Daniel Schlette, Philip Empl, Marco Caselli, Thomas Schreck, and Günther Pernul. 2024. Do You Play It by the Books? A Study on Incident Response Playbooks and Influencing Factors. In *2024 IEEE Symposium on Security and Privacy (SP).* 3625–3643. doi:10.1109/SP54263.2024.00060
[48] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal Policy Optimization Algorithms. *CoRR* (2017). arXiv:1707.06347 http://arxiv.org/abs/1707.06347 http://arxiv.org/abs/1707.06347.
[49] Arturo Servin and Daniel Kudenko. 2008. Multi-agent Reinforcement Learning for Intrusion Detection. In *Adaptive Agents and Multi-Agent Systems III. Adaptation and Multi-Agent Learning.*
[50] David Silver et al. 2016. Mastering the Game of Go with Deep Neural Networks and Tree Search. *Nature* 529, 7587 (Jan. 2016), 484–489. doi:10.1038/nature16961
[51] David Silver and Joel Veness. 2010. Monte-Carlo Planning in Large POMDPs. In *Advances in Neural Information Processing Systems*, Vol. 23.
[52] Rock Stevens, Daniel Votipka, Josiah Dykstra, Fernando Tomlinson, Erin Quartararo, Colin Ahern, and Michelle L. Mazurek. 2022. How Ready is Your Ready? Assessing the Usability of Incident Response Playbook Frameworks. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) *(CHI '22).* Association for Computing Machinery, New York, NY, USA, Article 589, 18 pages. doi:10.1145/3491102.3517559
[53] John N. Tsitsiklis and Benjamin van Roy. 1996. Feature-based methods for large scale dynamic programming. *Machine Learning* 22, 1 (01 Mar 1996), 59–94. doi:10.1007/BF00114724
[54] Sanyam Vyas, John Hannay, Andrew Bolton, and Professor Pete Burnap. 2023. Automated Cyber Defence: A Review. arXiv:2303.04926 [cs.CR] https://arxiv.org/abs/2303.04926, code: https://github.com/john-cardiff/-cyborg-cage-2.
[55] Stephen Walker and Nils Lid Hjort. 2001. On Bayesian consistency. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 63, 4 (2001), 811–821. doi:10.1111/1467-9868.00314
[56] Zelin Wan, Jin-Hee Cho, Mu Zhu, Ahmed H. Anwar, Charles A. Kamhoua, and Munindar P. Singh. 2022. Foureye: Defensive Deception Against Advanced Persistent Threats via Hypergame Theory. *IEEE Transactions on Network and Service Management* 19, 1 (2022), 112–129. doi:10.1109/TNSM.2021.3117698
[57] Wazuh Inc. 2022. Wazuh - The Open Source Security Platform. https://wazuh.com/
[58] Melody Wolk, Andy Applebaum, Camron Dennler, Patrick Dwyer, Marina Moskowitz, Harold Nguyen, Nicole Nichols, Nicole Park, Paul Rachwalski, Frank Rau, and Adrian Webster. 2022. Beyond CAGE: Investigating Generalization of Learned Autonomous Network Defense Policies. doi:10.48550/ARXIV.2211.15557
[59] Xin Xu and Tao Xie. 2005. A Reinforcement Learning Approach for Host-Based Intrusion Detection Using Sequences of System Calls. In *Advances in Intelligent Computing.*
[60] Huizhen Yu. 2006. *Approximate solution methods for partially observable markov and semi-markov decision processes.* Ph. D. Dissertation. Massachusetts Institute of Technology, USA. Advisor(s) Bertsekas, Dimitri.
[61] Huizhen Yu and Dimitri Bertsekas. 2004. Discretized approximations for POMDP with average cost. In *Proceedings of the 20th Conference on Uncertainty in Artificial Intelligence* (Banff, Canada) *(UAI '04).* AUAI Press, Arlington, Virginia, USA, 619–627.
[62] Lefeng Zhang, Tianqing Zhu, Farookh Khadeer Hussain, Dayong Ye, and Wanlei Zhou. 2022. Defend to Defeat: Limiting Information Leakage in Defending against Advanced Persistent Threats. *IEEE Transactions on Information Forensics and Security* (2022), 1–1. doi:10.1109/TIFS.2022.3229595
[63] Saman A. Zonouz, Himanshu Khurana, William H. Sanders, and Timothy M. Yardley. 2009. RRE: A game-theoretic intrusion Response and Recovery Engine. In *2009 IEEE/IFIP International Conference on Dependable Systems & Networks.* 439–448. doi:10.1109/DSN.2009.5270307
[64] Karl Johan Åström. 1965. Optimal control of Markov processes with incomplete state information. *J. Math. Anal. Appl.* 10, 1 (1965), 174–205. doi:10.1016/0022-247X(65)90154-X

A Experimental Setup

All computations are performed on an M2-ultra processor. The hyperparameters are listed in Table 3. Notation is explained in Table 4. We use the implementation of CARDIFF described in [54] and the implementation of C-POMCP described in [20]. For PPO, we use the STABLE-BASELINES implementation [44]. We set the hyperparameters for these methods to be the same as those used in [20]. We identify the dynamics of the quantized MDP through simulations. We solve the quantized MDP using value iteration.

Parameter(s)	*Values*
Convergence threshold of value iteration	0.1.
Number of particles M	50 [cf. (5)]
Discount factor γ	0.99 [cf. (2)]

Table 3: Hyperparameters.

B Proof of Proposition 1

Proposition 1 holds under the following two assumptions.

Notation(s)	*Description*
$\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{B}$	State, observation, action and belief spaces; cf. §3.
n	Number of states, i.e., $\mathcal{S} = \{1, 2, \ldots, n\}$; cf. §3.
$s_t, o_t, a_t, \mathbf{b}_t$	State, observation, action and belief at time t; cf. §3.
$p_{ss'}(a)$	Probability of the transition $s \to s'$ under action a; cf. §3.
$z(o \mid s)$	Probability of the observation o in state s; cf. §3.
$c(s, a)$	Cost in state s when taking action a; cf. §3.
Θ, ρ_t	Set of parameter vectors and conjecture distribution; cf. §3 and (6).
$\boldsymbol{\theta}, \overline{\boldsymbol{\theta}}$	True and conjectured parameter vector; cf. §3.
$\pi, \pi^\star, J^\star$	Strategy, optimal strategy, and optimal cost function; cf. (2).
$\mathbb{B}, \gamma$	The belief estimator and discount factor; cf. (1) and (2).
$\widehat{\mathbf{b}}_t, M$	Belief estimated through a particle filter and number of particles; cf. (5).
N	Number of system components in the illustrative example; cf. §5.
P, K	Probability measure and discrepancy function; cf. §4.2 and (8).
$K^\star, \Theta^\star$	The mininmal value of K and set of consistent conjectures; cf. Prop. 1 and (8).
$\hat{c}$	Cost function in the belief-MDP; cf. (3c).
$p_{\boldsymbol{\theta}}(\mathbf{b}' \mid \mathbf{b}, a)$	Probability of the transition in $\mathbf{b} \to \mathbf{b}'$ in the belief-MDP; cf. §3.
$\hat{p}_{\overline{\boldsymbol{\theta}}}(\mathbf{b}' \mid \mathbf{b}, a)$	Probability of the transition in $\mathbf{b} \to \mathbf{b}'$ in the (conjectured) quantized belief-MDP; cf. §4.3.
$\widehat{\mathcal{B}}, r$	Set of representative beliefs and quantization resolution; cf. (12).
$\Phi, \widehat{\mathbf{b}}$	Nearest-neighbor mapping and representative belief; cf. (13) and (12).
$\mu^\star, V^\star$	Optimal strategy and cost function in the quantized MDP; cf. (14).
$\overline{\pi}^\star, \overline{J}^\star$	Optimal strategy and cost function in the POMDP defined by $\overline{\boldsymbol{\theta}}$; cf. §4.3.
$\tilde{J}, \tilde{\pi}$	Cost function approximation and strategy approximation; cf. (14).
$S_{\widehat{\mathbf{b}}}$	Set of beliefs that are mapped to $\widehat{\mathbf{b}}$ in the quantization; cf. Prop. 3.

Table 4: Notation.

Assumption 1 (Well-defined prior and Bayesian learning).

(i) The set Θ is a compact subset of a Euclidean space.
(ii) The prior ρ_0 has full support, i.e, $\rho_0(\overline{\boldsymbol{\theta}}) > 0$ for all $\overline{\boldsymbol{\theta}} \in \Theta$.

Assumption 2 (Regularity conditions). *For any given observation $o \in \mathcal{O}$ and parameter vector $\boldsymbol{\theta} \in \Theta$,*

(1) The mapping $\mathbf{b} \mapsto \ln P(o \mid \boldsymbol{\theta}, \mathbf{b}, a)$ is Lipschitz w.r.t. the Wasserstein-1 distance, and the Lipschitz constant is independent of the observation o, the vector $\boldsymbol{\theta}$, and the action a.
(2) The mapping $\boldsymbol{\theta} \mapsto \ln P(o \mid \boldsymbol{\theta}, \mathbf{b}, a)$ is continuous and there exists an integrable function $g_{\mathbf{b},a}(o)$ for all beliefs $\mathbf{b} \in \mathcal{B}$ and actions $a \in \mathcal{A}$ such that $|\ln \frac{P(o|\boldsymbol{\theta},\mathbf{b},a)}{P(o|\overline{\boldsymbol{\theta}},\mathbf{b},a)}| \le g_{\mathbf{b},a}(o)$ for all parameter vectors $\overline{\boldsymbol{\theta}} \in \Theta$.

Due to page restrictions, we present only the main proof steps here. See our earlier work [21, Thm. 3] for technical details. To begin with, we invoke two lemmas from [21, Lemma 8, 9] that ensure the regularity of the belief space and the integrand in (9).

Lemma 1 (Compact Measure Space). *The belief space $\mathcal{B} \subset \mathbb{R}^n$ is compact with the Euclidean distance $d(\cdot,\cdot)$ and its corresponding Borel probability measure space $\Delta(\mathcal{B})$ is also compact with metric Wasserstein-p distance $d_{\mathcal{W}}(\cdot,\cdot)$.*

Lemma 2 (Continuity). *$\Delta K(\bar{\boldsymbol{\theta}}, \nu) \triangleq K(\bar{\boldsymbol{\theta}}, \nu) - K^\star_\Theta(\nu)$ is a continuous mapping defined over the product space $\mathcal{B} \times \Delta(\mathcal{B})$ with respect to the product metric of $d(\cdot,\cdot)$ and $d_{\mathcal{W}}(\cdot,\cdot)$.*

Finally, the following lemma clarifies the probability measure under which the almost-sure convergence holds [21, Lemma 6].

Lemma 3. *Any sequence of incident response strategies given by our method induces a well-defined probability measure over the sequence of historical states, partial observations, actions, and conjectures through the Ionescu-Tulcea extension [31].*

We now address the proof of Prop. 1. Given a conjecture $\overline{\boldsymbol{\theta}}$ and the true model $\boldsymbol{\theta}$, recursively applying (6) gives

$$\begin{aligned}\rho_{t+1}(\overline{\boldsymbol{\theta}}) &= \frac{\rho_0(\overline{\boldsymbol{\theta}}) \prod_{\tau=1}^t P(o_\tau \mid \overline{\boldsymbol{\theta}}, \mathbf{b}_{\tau-1}, a_{\tau-1})}{\int_\Theta \rho_0(\boldsymbol{\theta}') \prod_{\tau=1}^t P(o_\tau \mid \boldsymbol{\theta}', \mathbf{b}_{\tau-1}, a_{\tau-1}) \mathrm{d}\boldsymbol{\theta}'} \\ &= \frac{\rho_0(\overline{\boldsymbol{\theta}}) \exp\left(\ln\left(\prod_{\tau=1}^t \frac{P(o_\tau|\overline{\boldsymbol{\theta}},\mathbf{b}_{\tau-1},a_{\tau-1})}{P(o_\tau|\boldsymbol{\theta},\mathbf{b}_{\tau-1},a_{\tau-1})}\right)\right)}{\int_\Theta \rho_0(\boldsymbol{\theta}') \exp\left(\ln\left(\prod_{\tau=1}^t \frac{P(o_\tau|\boldsymbol{\theta}',\mathbf{b}_{\tau-1},a_{\tau-1})}{P(o_\tau|\boldsymbol{\theta},\mathbf{b}_{\tau-1},a_{\tau-1})}\right)\right) \mathrm{d}\boldsymbol{\theta}'} \\ &= \frac{\rho_0(\overline{\boldsymbol{\theta}}) \exp(-tZ_t(\overline{\boldsymbol{\theta}}))}{\int_\Theta \rho_0(\boldsymbol{\theta}') \exp(-tZ_t(\boldsymbol{\theta}')) \mathrm{d}\boldsymbol{\theta}'},\end{aligned}$$

where

$$Z_t(\overline{\boldsymbol{\theta}}) \triangleq t^{-1} \sum_{\tau=1}^t \ln\left(\frac{P(o_\tau \mid \boldsymbol{\theta}, \mathbf{b}_{\tau-1}, a_{\tau-1})}{P(o_\tau \mid \overline{\boldsymbol{\theta}}, \mathbf{b}_{\tau-1}, a_{\tau-1})}\right).$$

Plugging the expression above into the left-hand side of (9) yields

$$\frac{\int_\Theta \Delta K(\overline{\boldsymbol{\theta}}, \nu_t) \exp(-tZ_t(\overline{\boldsymbol{\theta}})) \rho_0(\overline{\boldsymbol{\theta}}) \mathrm{d}\overline{\boldsymbol{\theta}}}{\int_\Theta \rho_0(\boldsymbol{\theta}') \exp(-tZ_t(\boldsymbol{\theta}')) \mathrm{d}\boldsymbol{\theta}'}. \tag{18}$$

Given the structure of the numerator above, we can partition the set Θ into $\Theta^+_\epsilon \triangleq \{\overline{\boldsymbol{\theta}} : \Delta K(\overline{\boldsymbol{\theta}}, \nu_t) \ge \epsilon\}$ and $\Theta^-_{\epsilon/2} \triangleq \{\overline{\boldsymbol{\theta}} : \Delta K(\overline{\boldsymbol{\theta}}, \nu_t) \le \epsilon/2\}$, and the complement set of $\Theta^+_\epsilon \cup \Theta^-_{\epsilon/2}$ for any $\epsilon > 0$ and ν_t. Using this partitioning, (18) admits the following upper bound

$$\begin{aligned}&\frac{\int_\Theta \Delta K(\overline{\boldsymbol{\theta}}, \nu_t) \exp(-tZ_t(\overline{\boldsymbol{\theta}})) \rho_0(\overline{\boldsymbol{\theta}}) \mathrm{d}\overline{\boldsymbol{\theta}}}{\int_\Theta \rho_0(\boldsymbol{\theta}') \exp(-tZ_t(\boldsymbol{\theta}')) \mathrm{d}\boldsymbol{\theta}'} \\ &\quad\le \frac{\left(\int_{\Theta\setminus\Theta^+_\epsilon} + \int_{\Theta^+_\epsilon}\right) \Delta K(\overline{\boldsymbol{\theta}}, \nu_t) \exp(-tZ_t(\overline{\boldsymbol{\theta}})) \rho_0(\overline{\boldsymbol{\theta}}) \mathrm{d}\overline{\boldsymbol{\theta}}}{\int_\Theta \rho_0(\boldsymbol{\theta}') \exp(-tZ_t(\boldsymbol{\theta}')) \mathrm{d}\boldsymbol{\theta}'} \\ &\quad\le \epsilon + \underbrace{\frac{\int_{\Theta^+_\epsilon} \Delta K(\overline{\boldsymbol{\theta}}, \nu_t) \exp(-tZ_t(\overline{\boldsymbol{\theta}})) \rho_0(\overline{\boldsymbol{\theta}}) \mathrm{d}\overline{\boldsymbol{\theta}}}{\int_{\Theta^-_{\epsilon/2}} \rho_0(\boldsymbol{\theta}') \exp(-tZ_t(\boldsymbol{\theta}')) \mathrm{d}\boldsymbol{\theta}'}}_{(*)}.\end{aligned} \tag{19}$$

It suffices to prove that $(*)$ converges to zero for any $\epsilon > 0$.

Multiply both the numerator and denominator by $\exp(tK^\star_\Theta(\nu_t))$ in $(*)$, and we obtain

$$(*) = \frac{\int_{\Theta^+_\epsilon} \Delta K(\overline{\boldsymbol{\theta}}, \nu_t) \exp(-t(Z_t(\overline{\boldsymbol{\theta}}) - K^\star_\Theta(\nu_t))) \rho_0(\overline{\boldsymbol{\theta}}) \mathrm{d}\overline{\boldsymbol{\theta}}}{\int_{\Theta^-_{\epsilon/2}} \rho_0(\boldsymbol{\theta}') \exp(-t(Z_t(\boldsymbol{\theta}') - K^\star_\Theta(\nu_t))) \mathrm{d}\boldsymbol{\theta}'}.$$

According to [21, Lemma 7], $\lim_{t\to\infty} |Z_t(\overline{\boldsymbol{\theta}}) - K(\overline{\boldsymbol{\theta}}, \nu_t)| = 0$, almost surely, which implies that asymptotically, $Z_t(\overline{\boldsymbol{\theta}}) - K^\star_\Theta(\nu_t)$ is equivalent to $K(\overline{\boldsymbol{\theta}}, \nu_t) - K^\star_\Theta(\nu_t)$, and hence,

$$\begin{aligned}(*) &= \frac{\int_{\Theta^+_\epsilon} \Delta K(\overline{\boldsymbol{\theta}}, \nu_t) \exp(-t\Delta K(\overline{\boldsymbol{\theta}}, \nu_t)) \rho_0(\overline{\boldsymbol{\theta}}) \mathrm{d}\overline{\boldsymbol{\theta}}}{\int_{\Theta^-_{\epsilon/2}} \exp(-t\Delta K(\boldsymbol{\theta}', \nu_t)) \rho_0(\boldsymbol{\theta}') \mathrm{d}\boldsymbol{\theta}'} \\ &\le \frac{\epsilon e^{-t\epsilon}}{\int_{\Theta^-_{\epsilon/2}} e^{-\epsilon t/2} \rho_0(\boldsymbol{\theta}') \mathrm{d}\boldsymbol{\theta}'}.\end{aligned}$$

Therefore, to prove that the upper bound of the left-hand side of (9) vanishes, we need to show that $\rho_0(\Theta^-_{\epsilon/2}) \triangleq \int_{\Theta^-_{\epsilon/2}} \rho_0(\boldsymbol{\theta}') \mathrm{d}\boldsymbol{\theta}'$ is strictly greater than zero for every t, for which the compactness result in Lemma 1 becomes helpful.

Compactness of the parameter set Θ and the continuity of $\Delta K(\overline{\theta}, \nu)$ imply its uniform continuity. Berge's maximum theorem implies that $\Theta^\star(\nu)$ is non-empty [1, Thm. 17.31]. Therefore, for any $\overline{\theta}_v \in \Theta^\star(\nu)$, there exist $\overline{\theta} \in \Theta$, $\nu' \in \Delta(\mathcal{B})$, and δ_m such that when $d(\overline{\theta}_\nu, \overline{\theta}') < \delta_m$ and $d_{\mathcal{W}}(\nu, \nu') < \delta_m$, $\Delta K(\overline{\theta}', \nu') = \Delta K(\overline{\theta}', \nu') - \Delta K(\overline{\theta}_\nu, \nu) \leq m$, where the equality follows because $\overline{\theta}_\nu \in \Theta^\star(\nu)$ implies $\Delta K(\overline{\theta}_\nu, \nu) = 0$. As a result, for any $\nu \in \Delta(\mathcal{B})$ and $\nu' \in B(\nu, \delta_m) \triangleq \{\nu' \mid d_{\mathcal{W}}(\nu, \nu') < \delta_m\}$,

$$\underbrace{\{\overline{\theta}' \mid d(\overline{\theta}', \overline{\theta}_\nu) < \delta_m\}}_{\Theta_\nu(\delta_m)} \subset \underbrace{\{\overline{\theta}' \mid \Delta K(\overline{\theta}', \nu') \leq m\}}_{\Theta_{\nu'}(m)}.$$

Thus, for any ν and $\nu' \in B(\nu, \delta_m)$, $\rho_0(\Theta_{\nu'}(m)) \geq \rho_0(\delta_m)) > 0$, where the strict inequality follows because ρ_0 has full support (Assumption 1).

The set $\{B(\nu, \delta_m)\}_{\nu \in \Delta(\mathcal{B})}$ forms an open cover for a compact space, implying that there exists a finite subcover $\{B(\nu_i, \delta_m)\}_{i=1}^M$. As a consequence, $\nu' \in \Delta(\mathcal{B})$ belongs to some Wasserstein ball $B(\nu_i, \delta_m)$. Let $r \triangleq \min_i \rho_0(\Theta_{\nu_i}(\delta_m)) > 0$. We obtain $\rho_0(\Theta_{\nu'}(m)) \geq \rho_0(\Theta_{\nu_i}(\delta_m)) \geq r$, which yields $\rho_0(\Theta^-_{\epsilon/2}) \geq r > 0$ with $m = \epsilon/2$. □

C Proof of Proposition 2

The proof follows the same chain of reasoning as the proof of the simulation lemma in [32]. We start by expanding the difference $|\overline{J}^\star(\mathbf{b}) - J^\star(\mathbf{b})|$ as follows.

$$\begin{aligned}
|\overline{J}^\star(\mathbf{b}) - J^\star(\mathbf{b})| &= \Bigg| \hat{c}(\mathbf{b}, a) + \gamma \sum_{\mathbf{b}' \in \mathcal{B}} p_{\overline{\theta}}(\mathbf{b}' \mid \mathbf{b}, a) \overline{J}^\star(\mathbf{b}') - \Bigg(\\
&\qquad \hat{c}(\mathbf{b}, a) + \gamma \sum_{\mathbf{b}' \in \mathcal{B}} p_\theta(\mathbf{b}' \mid \mathbf{b}, a) J^\star(\mathbf{b}') \Bigg) \Bigg| \\
&= \left| \gamma \sum_{\mathbf{b}' \in \mathcal{B}} p_{\overline{\theta}}(\mathbf{b}' \mid \mathbf{b}, a) \overline{J}^\star(\mathbf{b}') - \gamma \sum_{\mathbf{b}' \in \mathcal{B}} p_\theta(\mathbf{b}' \mid \mathbf{b}, a) J^\star(\mathbf{b}') \right| \\
&= \Bigg| \gamma \sum_{\mathbf{b}' \in \mathcal{B}} p_{\overline{\theta}}(\mathbf{b}' \mid \mathbf{b}, a) \overline{J}^\star(\mathbf{b}') - \gamma \sum_{\mathbf{b}' \in \mathcal{B}} p_\theta(\mathbf{b}' \mid \mathbf{b}, a) J^\star(\mathbf{b}') + \\
&\quad \gamma \sum_{\mathbf{b}' \in \mathcal{B}} p_{\overline{\theta}}(\mathbf{b}' \mid \mathbf{b}, a) J^\star(\mathbf{b}') - \gamma \sum_{\mathbf{b}' \in \mathcal{B}} p_{\overline{\theta}}(\mathbf{b}' \mid \mathbf{b}, a) J^\star(\mathbf{b}') \Bigg| \\
&= \Bigg| \gamma \sum_{\mathbf{b}' \in \mathcal{B}} p_{\overline{\theta}}(\mathbf{b}' \mid \mathbf{b}, a) \left(\overline{J}^\star(\mathbf{b}') - J^\star(\mathbf{b}') \right) + \\
&\quad \gamma \sum_{\mathbf{b}' \in \mathcal{B}} \left(p_{\overline{\theta}}(\mathbf{b}' \mid \mathbf{b}, a) - p_\theta(\mathbf{b}' \mid \mathbf{b}, a) \right) J^\star(\mathbf{b}') \Bigg| \\
&\leq \gamma \|\overline{J}^\star - J^\star\|_\infty + \left| \gamma \sum_{\mathbf{b}' \in \mathcal{B}} \left(p_{\overline{\theta}}(\mathbf{b}' \mid \mathbf{b}, a) - p_\theta(\mathbf{b}' \mid \mathbf{b}, a) \right) J^\star(\mathbf{b}') \right| \\
&\overset{(a)}{\leq} \gamma \|\overline{J}^\star - J^\star\|_\infty + \gamma \sum_{\mathbf{b}' \in \mathcal{B}} \left| \left(p_{\overline{\theta}}(\mathbf{b}' \mid \mathbf{b}, a) - p_\theta(\mathbf{b}' \mid \mathbf{b}, a) \right) \right| \frac{c_{\text{MAX}}}{1 - \gamma} \\
&\leq \gamma \|\overline{J}^\star - J^\star\|_\infty + \frac{\gamma \alpha c_{\text{MAX}}}{1 - \gamma},
\end{aligned}$$

where $\hat{c}(\mathbf{b}, a)$ is defined in (3c) and (a) follows because $|J^\star(\mathbf{b})| \leq \sum_{t=0}^\infty \gamma^t c_{\text{MAX}} = \frac{c_{\text{MAX}}}{1-\gamma}$ and the fact that $|ab| = |a||b|$ (we use the triangle inequality to move the absolute value inside the sum). Since this upper bound holds for any belief state $\mathbf{b}$, we have

$$\begin{aligned}
&\|\overline{J}^\star - J^\star\|_\infty \leq \gamma \|\overline{J}^\star - J^\star\|_\infty + \frac{\gamma \alpha c_{\text{MAX}}}{1 - \gamma} \\
\implies &\|\overline{J}^\star - J^\star\|_\infty - \gamma \|\overline{J}^\star - J^\star\|_\infty \leq \frac{\gamma \alpha c_{\text{MAX}}}{1 - \gamma} \\
\implies &(1 - \gamma) \|\overline{J}^\star - J^\star\|_\infty \leq \frac{\gamma \alpha c_{\text{MAX}}}{1 - \gamma} \\
\implies &\|\overline{J}^\star - J^\star\|_\infty \leq \frac{\gamma \alpha c_{\text{MAX}}}{(1 - \gamma)^2}. \quad \square
\end{aligned}$$

D Proof of Proposition 3

The result expressed in Prop. 3 was originally proven by Tsitsiklis and van Roy in [53, Thm. 1], and later generalized by Li et al. [39, Prop. 3]. Variants of this proof are also presented by Bertsekas in [10–12]. As this result is well established, we omit the proof.

E Proof of Proposition 4

Our proof is based on the arguments outlined by Hammar et al. in [22, Prop. 2]. See also the proofs by Saldi et al. in [46, Thm. 2.2] and Yu and Bertsekas [61, Thm. 1] for extensions to non-finite POMDPs and POMDPs with the average-cost criterion.

It can be shown that the (conjectured) optimal cost function $\overline{J}^\star : \mathcal{B} \mapsto \mathbb{R}$ is uniformly continuous; see e.g., [60, Prop. 2.1]. Fix an arbitrary scalar $\omega > 0$. By uniform continuity, there exists a scalar $\delta > 0$ such that

$$\|\mathbf{b} - \mathbf{b}'\|_\infty < \delta \implies |\overline{J}^\star(\mathbf{b}) - \overline{J}^\star(\mathbf{b}')| < \omega, \tag{20}$$

for all $\mathbf{b}, \mathbf{b}' \in \mathcal{B}$. The quantization in (12) partitions $\mathcal{B}$ into grid cells $S_{\tilde{\mathbf{b}}}$ with resolution $r \geq 1$; cf. (15). Further, (13) implies that if $\mathbf{b} \in S_{\tilde{\mathbf{b}}}$, then

$$\|\mathbf{b} - \tilde{\mathbf{b}}\|_\infty = \min_{\tilde{\mathbf{b}}' \in \tilde{\mathcal{B}}} \|\mathbf{b} - \tilde{\mathbf{b}}'\|_\infty.$$

Because each belief coordinate $\mathbf{b}(s)$ lies in $[0, 1]$ and each representative belief coordinate $\tilde{\mathbf{b}}(s)$ equals $\frac{\beta_s}{r}$ for some $\beta_s \in \{0, \ldots, r\}$ [cf. (12)], we have

$$\max_{\mathbf{b}, \mathbf{b}' \in S_{\tilde{\mathbf{b}}}} \|\mathbf{b} - \mathbf{b}'\|_\infty \leq \frac{2n}{r}, \qquad \text{for every } \tilde{\mathbf{b}} \in \tilde{\mathcal{B}}.$$

Choose any r such that $\frac{1}{r} < \delta$. By (20), we have

$$|\overline{J}^\star(\mathbf{b}) - \overline{J}^\star(\mathbf{b}')| < \omega, \quad \text{for all } \mathbf{b}, \mathbf{b}' \in S_{\tilde{\mathbf{b}}},\ \tilde{\mathbf{b}} \in \tilde{\mathcal{B}}.$$

Because $\omega > 0$ is arbitrary and there exists a large enough r such that $\frac{1}{r} < \delta$ for any $\delta > 0$, we have

$$\lim_{r \to \infty} \max_{\tilde{\mathbf{b}} \in \tilde{\mathcal{B}}} \max_{\mathbf{b}, \mathbf{b}' \in S_{\tilde{\mathbf{b}}}} |\overline{J}^\star(\mathbf{b}) - \overline{J}^\star(\mathbf{b}')| = 0.$$

Hence the constant ϵ in Prop. 3 diminishes as $r \to \infty$. Invoking the error bound in Prop. 3 completes the proof. □